



Deckblatt Auftragsdokumentation

Coverage Documentation

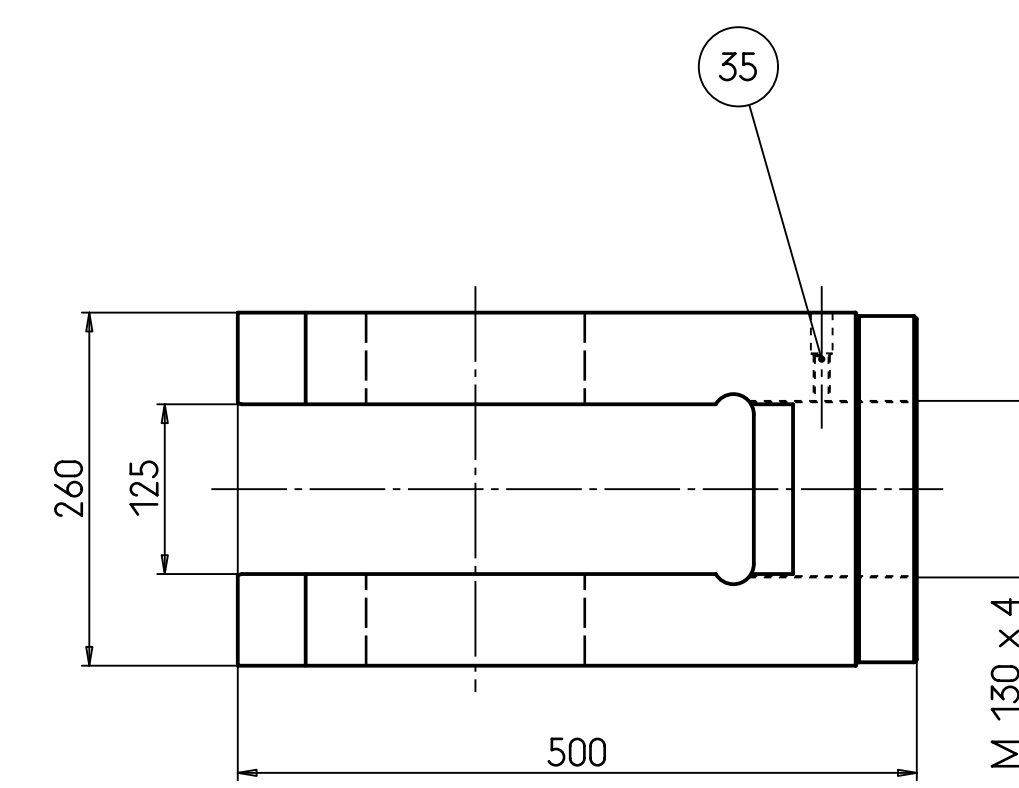
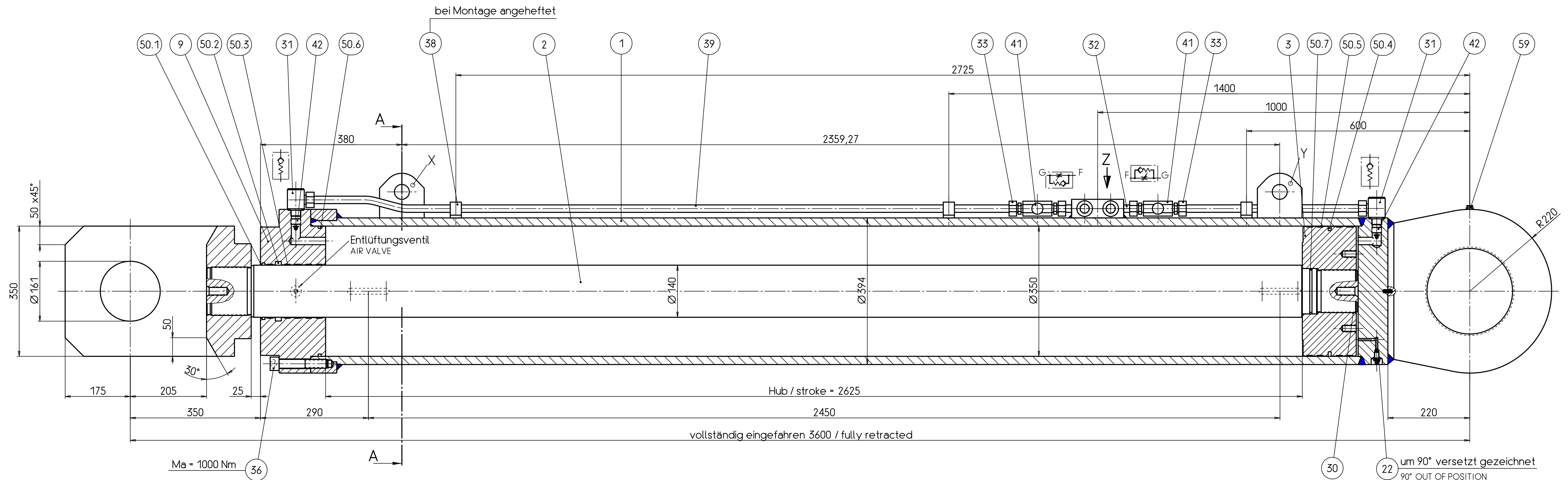
Coverture Documentation

Gegenstand OBJECT OBJET	:	Ausschwenkzylinder SWINGING DEVICE CYL. VÉRIN À BASCULE
Abmessung MEASURE MÉTRAGE	:	D 350/140x2625 Hub / stroke / course
Loesche Bestell-Nr. LOESCHE ORDER-NO. LOESCHE COMMANDE-NO.	:	4501.7985
Kennwort KEYWORD REFERENCE	:	Narayangonj
Hersteller Auftrags-Nr. MANUFACTURE ORDER-NO. REF.NO.DU MANUFACTEUR	:	107237
Erstellungsdatum MANUFACTURING DATE DATE DE MANUFACTURATION	:	2004

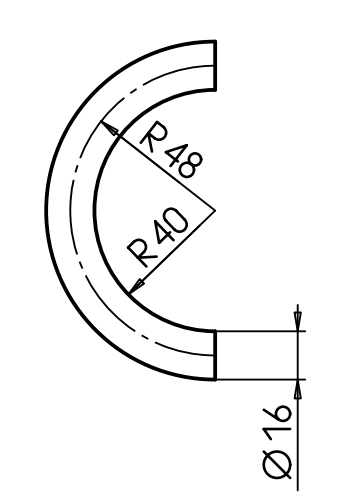
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- 3. Montage, Inbetriebnahme u. Wartung**
ASSEMBLY, START-UP AND MAINTENANCE
INSTRUCTION DE MONTAGE

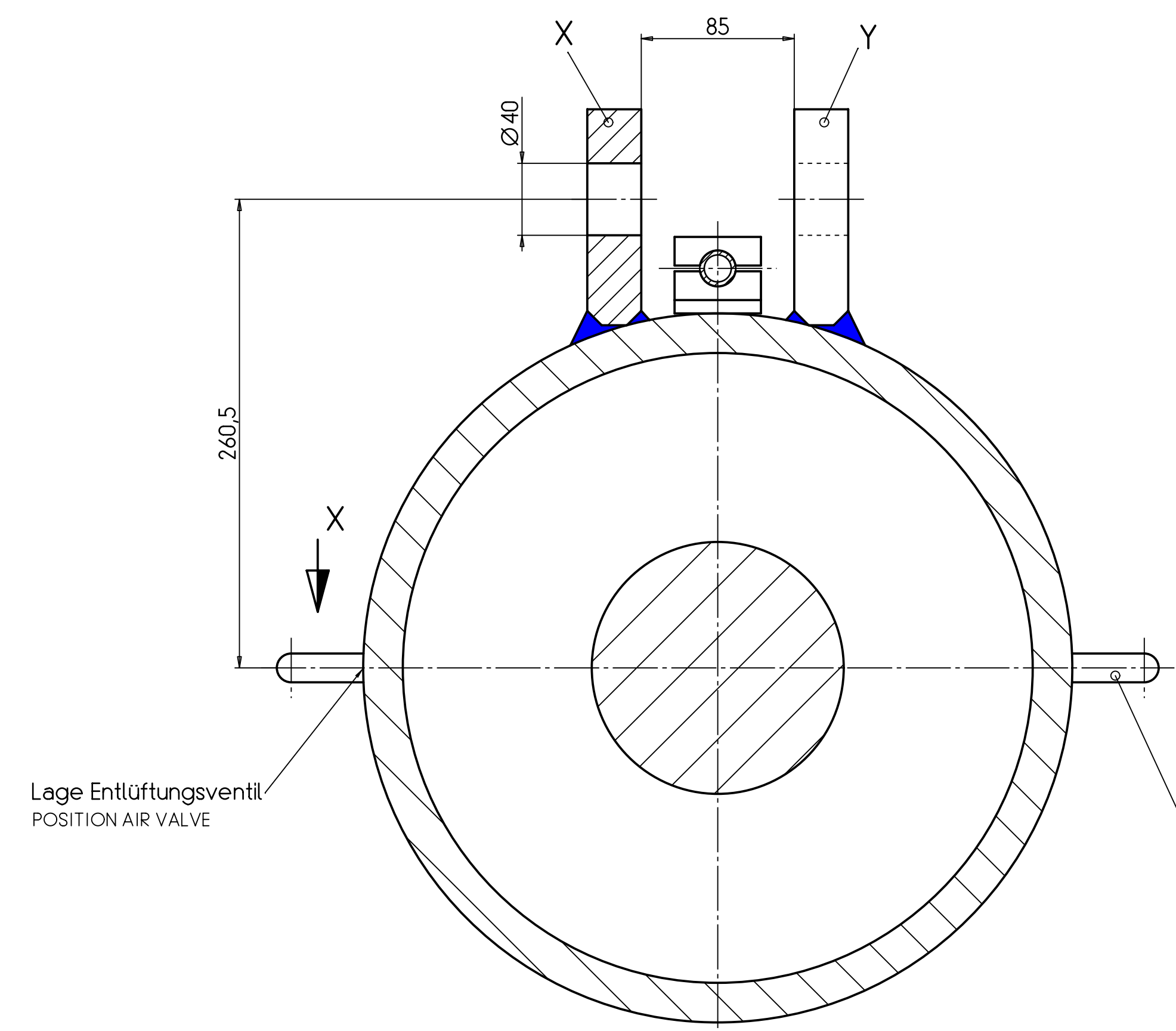
Ersatzteilliste / Spare parts list							
Bezeichnung :			Ausschwenkzylinder / Swinging device cyl.				
Type							
Abmessung :			D 350/140 - 2625 Hub / stroke		Kennwort:	Narayangonj	
Meassure					Key Word		
RUHFUS-Zeichnungs Nr.:			50-1663-10				
RUHFUS-Drawing-No.							
Pos.	St.	Bezeichnung	Zeichnungs-Nr.	Abmessung	Qualität	DIN / EN	Bemerkungen
Pos.	Piec.	Name	Drawing-No.	Measure	Quality	DIN / EN	Remarks
1	1	Zylinderrohr kpl.	1453.01.00.00.00.2/0				
		Cyl. Tube assy.					
2	1	Kolbenstange	1453.02.00.00.00.2/0				
		Piston Rod					
3	1	Kolben	1453.03.00.00.00.2/0				
		Piston					
9	1	Zylinderkopf	1453.09.00.00.00.1/0				
		Cylinder head					
16	1	Gabelkopf	1663.16.00.00.00.1/0				
		Rod clevis					
22	2	Entlüfterventil		M12x1,5x43			ATE
		Air valve					
30	2	Gewindestift		M8x12		916	
		Hexagon socket screw					
31	4	Schwenkverschraubung	DSVW 20-PSR				
		Banjos					
32	2	Gerader Einschraubstutzen	EVGE 20-PSR-ED				
		Male stud adaptors					
33	2	Gereade E.-verschrbg.	GE 20-PSR-ED				
		Male stud coupl.					
35	1	Gewindestift		M12x45		913	
		Hexagon socket screw					
36	22	Zylinderschraube		M24x130	10.9	912	
		Screw					
38	3	Rohrschelle	RAPR -320				
		Tube clamp					
39	1	Präz-Rohr		D 20x2,5x2900	St 35.4	2391	
		Hose					
41	2	Rückschlagventil	RD 4				
		Throtteling check valve					
42	2	Leitungsbruchsicherung	LB4-C				
		Pipe valve					
50	1	Dichtsatz kpl.	1453.EDS				
		Sealing set cpl.					
50.1	1	Abstreifer		D 140/152x12	PTFE-Bz/NBR		
		Scraper ring					
50.2	1	Stangendichtung		D 140/160x14,6	NBR / Gewebe		
		Chevron sael					
50.3	5	Stangenführungsring		D 140/145x15	Hartgewebe		
		Piston wear ring					
50.4	1	Kolbendichtung		D 350/325,5x8,1			
		Piston seal					
50.5	4	Kolbenführungsring		D 350/345x15	Hartgewebe		
		Piston wear ring					
50.6	1	O-Ring		D 336x7	NBR 70		
		O-seal					
50.7	1	O-Ring		D 112x5	NBR 70		
		O-seal					



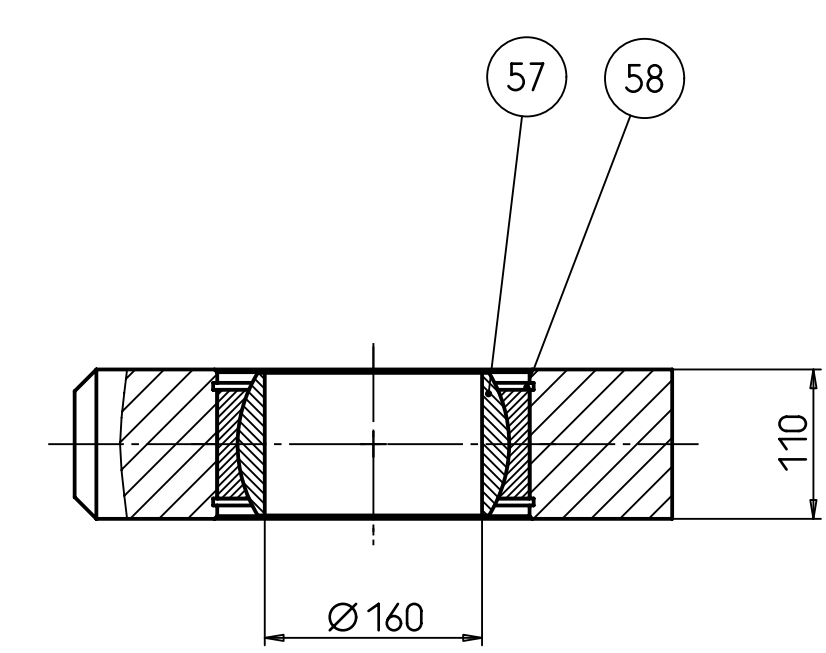
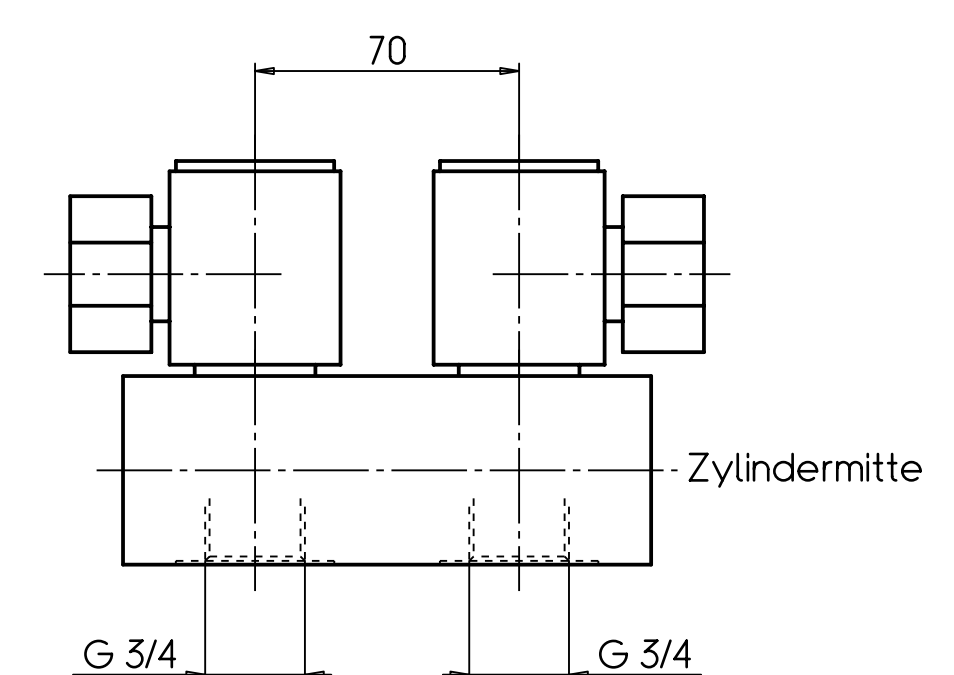
DETAIL
Einzelheit X



SECTION
Schnitt A-A
M 1: 2,5



VIEW
Ansicht Z
M 1: 2



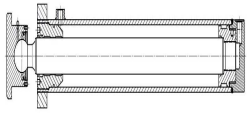
Achtung:
Ösen dienen nicht zum Transport des Zylinders

Betriebsdruck 160 bar
WORKING PRESSURE
Prüfdruck 240 bar
TEST PRESSURE

Produktionsfreigabe
Kom-Nr.
Name: Datum

Urheberschutz nach DIN 34 vorbehalten Copyright reserved RUHFUS Systemhydraulik GmbH		CAD-Zeichnung		Maßstab * / *	Gewicht in kg : c8. 1700
Datum 03.06.2004 Bearb. A.Schwirn Geprf. Norm		Loesche-Nr. M907002-13-2		Ausschwenkzylinder SWINGING DEVICE CYL. D 350/140 - 2625 Hub/stroke	
RUHFUS SYSTEMHYDRAULIK		50-1663-1-0		Blatt 1	
Zust.	Mittlung	Datum	Name	Bl.	

Abw. für Längenmaße und Rundungen nach DIN 7168 Teil 1 - mittel.						Allg. Toleranzen nach DIN 7168 Teil 2 für Form - und Lage	
bis 6	über 6 bis 50	über 50 bis 120	über 120 bis 400	über 400 bis 1000	über 1000 bis 2000	Genauigkeitsgrad	
± 0,1	± 0,2	± 0,3	± 0,5	± 0,8	± 1,2		

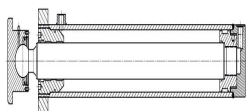


Assembly,
Start-up and Maintenance
of Hydraulic Cylinders

Data Sheet No.
HY-3910-4
Issued: 11/2002

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2. Mounting and Assembly
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4. Maintenance
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Assembly, Start-up and Maintenance of Hydraulic Cylinders

Data Sheet No.
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1. General Description

Before starting up the hydraulic cylinder consult the following standards:

- | | |
|----------------------|----------------------|
| - German standards | DIN 24346, para. 5.2 |
| - ISO standard | ISO 4413, para. 6.2 |
| - European standards | EN 982, para. 5.3.2 |

2. Mounting and Assembly

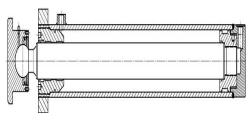
- Cleanliness:

- When mounting the cylinder and in the vicinity
- of pipes and containers all traces of soiling, scales, sand, chips, etc. are to be cleaned before starting assembly
- hot-bent or welded pipes are to be finally pickled, cleaned and oiled.
- When cleaning use only non-linting cloths or special paper
- Do not use sealants such as hemp, putty or sealing tape

3. Start-up

3.1 Pressure Liquid

- Observe the set pressure and temperature ranges.
- Check whether the pressure liquid in the system matches the pressure liquid recommended for the hydraulic cylinder.
- The maximum temperature recommended by the oil manufacturer must not be exceeded.
In order to assure the system's constant operating performance, it is recommended to keep a constant pressure liquid temperature ($\pm 5^{\circ}\text{C}$).



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- As a rule mineral oils (HL, HLP) as per DIN 51524 can be used
Other pressure liquids only at request.

3.2 Filtering

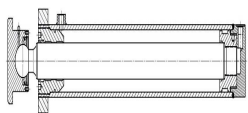
- A reliable form of filtering can increase the hydraulic cylinder's service life. Follow the recommendations for the maximum permissible fouling factor for the pressure liquid as per NAS 1638 / ISO DIS 4406.
- Filters with electric soiling indicators are to be preferred.
- When changing the filter make sure to provide for the cleanliness.

3.3 Bleeding

- At idle run pressure loosen the bleeder screw or the bottom- or rod-side screw.
- Let the air escape
- Tighten the bleeder screw or the other relevant screws only when the outflowing oil is bubble-free.

4. Maintenance

- Generally speaking, the hydraulic cylinder does not require any maintenance; make sure to lubricate the various bearing such as the swivel pins as well as the spindle carriers and ball-and-socket joints.
- Check for the leak-proof closure at brief intervals
- The motion seals are parts subject to wearing. Should the internal or external leakage become too great, we recommend that you change the seals or send the hydraulic cylinder to our factory so that the guides and the other parts can be checked during the changing of the seals.



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5. Usage limitations for sealing and guide elements

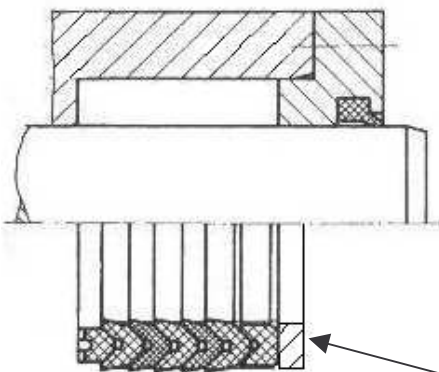
- Temperature range -20°C to $+80^{\circ}\text{C}$
- Max. sliding speeds
 - Chevron packings; grooved rings : 0.5 m/sec
 - Glydring seals : up to 15 m/sec
 - Scraper : 4 – 15 m/sec.
(manufacturer-specific)
 - Guide elements : min. 1 m/sec or higher
(manufacturer-specific)
- You can find the various pressure ratings and proof pressures in the assembly drawings.

6. Assembly of seals / scrapers / guide elements

6.1 Rod seals

6.1.1 Chevron packings

Chevron packing in an open chamber

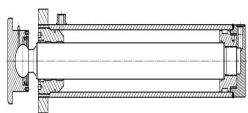


Chevron packing in a closed
chamber with support ring

Before mounting all of the single parts of the seal unit are to be greased. You can use mineral greases as long as these prove to be suitable for seals.

With rod diameters of 140 mm or more the chevron packings can be mounted in closed chambers. To compensate for the larger chamber a support ring is to be inserted.

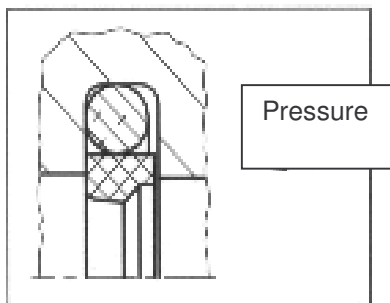
Support ring with closed slot



Assembly, Start-up and Maintenance of Hydraulic Cylinders

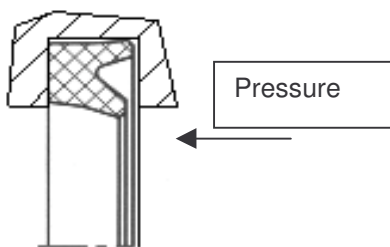
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6.1.2 Glydrings



When mounting the Glydring seals the O-ring is to be positioned in the mounting groove and then the profile is deformed to assume a kidney-like shape and snaps into place. Make sure that the profile ring has no sharp bends and that the sealing edge is placed correctly according to the direction of the pressure exerted.

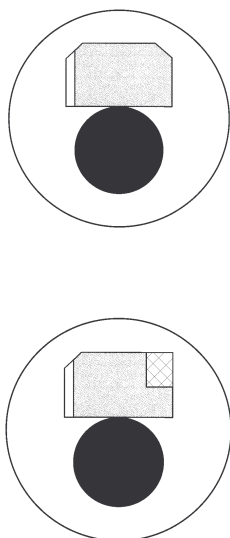
6.1.3 Grooved rings



The grooved rings can be mounted by hand in the recessed grooves. Make sure that the sealing lip is not damaged and that the sealing lip is placed correctly according to the direction of the pressure exerted.

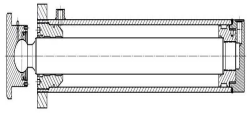
6.2 Piston Packings

6.2.1 Glydrings



Glydring piston packings can be mounted on solid pistons. The O-ring is mounted first and then the profile ring is pushed on. Warming the profile ring in hot water or with a jet of hot air at about 80 °C – 100 °C makes the mounting easier. The gauging of the seal can be performed with the relevant cylinder tube. This operation must be performed immediately after fitting.

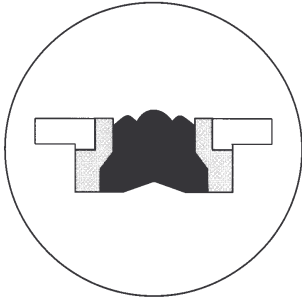
A special seal is the Glydring with gusset. The gusset is an additional, slotted support ring made of heat-resistant elastomer. When using this solution, follow the instructions provided in the assembly drawings.



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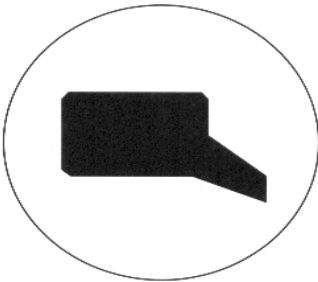
6.2.2 Compact Seals



Multi-unit piston packings can be snapped on for the most part with the aid of tools. First the rubber elastic contact element is to be snapped on. Then the slotted support rings and welded-angle rings are to be inserted.

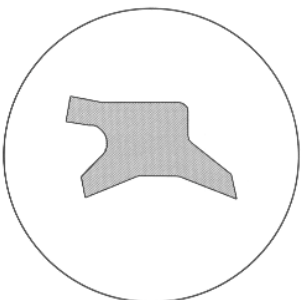
6.3 Scraper

6.3.1 Single-acting version

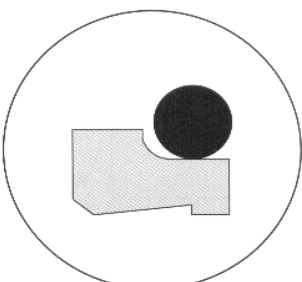


Generally speaking, the scrapers can be mounted quickly and easily thanks to their capacity to assume a kidney-like shape. The scrapers with metal retainers are pressed into an axially accessible housing space.

6.3.2 Double-acting version



One-part version



Two-part version

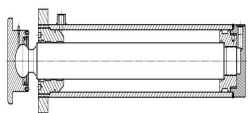
Double scrapers come into two different versions:

- One-part version (double-lip)
- Two-part version.

The thorough assembly of the equipment is the prerequisite for its proper functioning. Generally speaking, the scrapers can be mounted quickly and easily thanks to its capacity to assume a kidney-like shape.

The two-part double scrapers are composed of an oil control ring and an O-ring. Depending on the manufacturer even 2 O-rings can be used as the pretensioning element.

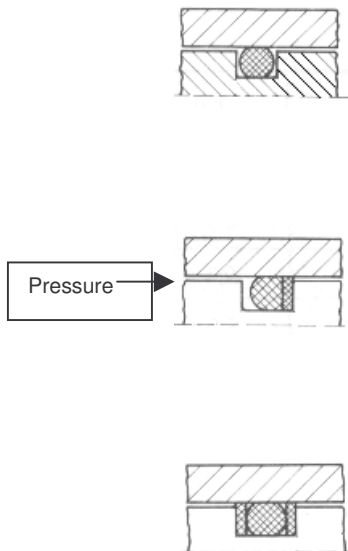
Follow carefully the assembly instructions in the drawings.



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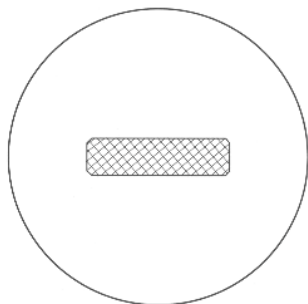
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6.4 O-rings



The O-rings must not be pulled over sharp edges or exposed shafts, threads, grooves, etc. It is recommended to cover the sharp spaces during assembly to prevent their damaging. The O-rings must not be twisted or drilled in the housing groove. They can be slightly stretched while being mounted. However, after mounting they must be left to rest to allow them to resume their original shape. To prevent extruding through the slot additional support rings can be mounted. Follow the instructions in the assembly drawing.

6.5 Guide Rings

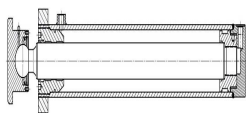


Their purpose is that of guiding the piston and the rod, preventing metallic friction between the sliding components and supporting the resulting lateral forces. Follow the assembly instructions in the drawings.

7. Transportation and Storage

When handling the cylinder make sure that

- the cylinder is always completely run in to prevent the damaging of the piston rod
- Avoid knocking against the piston rod guide and the piston during handling to prevent the damaging of the guide elements and the piston rod.



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- c) The hydraulic connections are to be sealed with screw plugs or other suitable caps or covers with O-rings to prevent the infiltration of soiling

In case of storage for long periods of time all of the bare surfaces are to be coated with Tectyl.

In case of storage periods lasting more than 6 months the hydraulic cylinder is to be filled with a preservative oil and seal.

We recommend for instance RENOLIN MR 15 by Fuchs Lubritech GmbH.

When starting the cylinder there is no need to remove this oil from the cylinder chamber. This oil blends perfectly with all mineral oils as per DIN 51124.

The hydraulic cylinder is to be kept in a dry and dustless room free from corrosive substances and dampness.

If possible, the hydraulic cylinders are to be kept in a vertical position in case of long storage periods.

8. Safety Specifications

The safety requirements envisaged in standard EN 982, para. 5.3.2, in its most recent valid version are to be applied.